Client Project Scope - Arindam + Dip

Scope Statement

Itinera is a clean and lightweight mobile application that provides tourists with a personalised travel itinerary given their time schedule, cost budget and preferred attractions. We aim to deliver a travel itinerary that maximises the tourist's utility while providing interactive features such as allowing the user to edit their budget, time schedule and view POI details. We will exclude features such as adding, removing and reordering POIs. The primary goal of Itinera is to assist tourists with the complexity of navigating unfamiliar destinations given limited resources and areas of interest.

Team Project Proposal

Computer Science Project

COMP3900

25T3

Team Bananas

Project 114

A Fast, Scalable, And Optimal Personalised Travel Itinerary Planning System.

4 October 2025

3900-H13A-BANANA			
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Background

Problem Definition and Impact

Tourism is a substantial and growing part of the economy, and travellers increasingly expect personalised, feasible day plans that fit their time and cost budgets while respecting opening hours and other practical constraints. In Australia, tourism contributed $78.1 billion (approximately 2.9% of GDP) in 2023–24 (Australian Bureau of Statistics 2024). Internationally, tourist arrivals fully recovered in 2024 and grew a further 5% in the first half of 2025 compared with the same period in 2024, indicating sustained demand for reliable trip-planning support (UN Tourism 2025a; UN Tourism 2025b). Inbound spending has also rebounded strongly; for example, international overnight visitor expenditure in Australia reached a record $32.9 billion in 2024 (Tourism Research Australia 2024). These indicators together show that more travellers are planning dense city days and need tools that handle real-world scheduling and budget constraints effectively.

Our team is tasked with constructing a day plan (itinerary), which is a sequence of points of interest (POIs) with arrival and departure times in which each POI is visited at most once, all within traveller-specified time and cost budgets and opening-hour constraints. Constructing such an itinerary is challenging because the set of plausible POI combinations and visit orders grows rapidly, and many appealing sequences become infeasible once opening hours and travel times are considered. This task aligns with the Orienteering Problem (OP) and its time-window variants, which are combinatorial and NP-hard; the addition of time windows further increases the difficulty of finding feasible, high-quality solutions at realistic city scales (Vansteenwegen, Souffriau & Van Oudheusden 2011; Gunawan, Lau & Vansteenwegen 2016). In practice, this complexity explains why pre-defined itineraries rarely match a traveller’s unique constraints and why manual planning is slow and error-prone.

Constraints	Practical Effect on the Traveller
Time budget limits the achievable set of POIs.	A fixed day means only a subset of attractive POIs can be included, and every minute spent travelling reduces time available on site.
Cost budget forces value–spend trade-offs.	Entrance fees and extras push some POIs out of reach and require prioritisation of higher-utility stops within a fixed monetary limit.
Opening hours create strict time windows.	A sequence that appears promising can become infeasible if a museum opens late or a garden closes early.
Inter-POI travel times make the visit order critical.	The same set of POIs can be feasible or infeasible depending on their order because long links between distant POIs can consume the time budget and break later time-window constraints.
Heterogeneous user interests demand personalisation.	Two travellers with different preferences (for example, art versus parks) derive different utility from the same city, as a result, generic pre-defined itineraries rarely fit individual tastes or constraints.

Proposed Solution

Our proposed solution is a lightweight mobile app *Itinera* which provides a personalised itinerary that respects the traveller's available time and overall budget. Itinera allows them to generate and review a realistic schedule without managing technical details or complex settings.

1. **Feasible, preference-aware planning**
   The system produces an ordered day plan with clear arrival and departure times for each point of interest (POI). It balances the traveller's indicated preferences, for example galleries versus parks, with practical constraints such as indicative visit durations and estimated travel time between locations.
2. **Clean, lightweight, and mobile-centric interface**
   The application focuses on a simple input flow, the traveller sets a day window, an overall cost budget, and high-level interest categories. The resulting itinerary is a simple map that highlights arrival and departure times, total duration and cost.
3. **Transparent information and controlled adjustments**
   Each POI includes concise details such as category, indicative time on site, entry fee, and opening hours. Providing this context helps users understand why a certain attraction appears in the plan and how it contributes to the overall day.

Review of Existing Systems

TripIt

Overview

TripIt is a mobile itinerary manager that consolidates a traveller's bookings into a single master schedule. Users forward confirmation emails or enable inbox synchronisation, and the service parses flights, accommodation and transport into a structured timeline on phone and web, reducing the need to juggle multiple documents and calendars (TripIt Help, 2025). The app is optimised for reliability during travel, including access to stored trip details without connectivity provided the user remains signed in, which supports use in areas with poor reception and on aeroplanes (TripIt Help, 2023). A premium tier, TripIt Pro, adds real-time operational information for flights, such as status alerts, terminal and gate reminders and interactive airport maps, improving situational awareness on travel days and reducing missed updates that can cause delays and stress (TripIt Pro, 2025). In summary, TripIt's core value proposition is administrative, it centralises logistics that a traveller has already booked, and it presents them clearly in one place so the user can focus attention on execution

Strengths

TripIt's automation significantly lowers data entry burden. Forwarded emails and inbox synchronisation create itineraries with minimal effort and maintain consistency across devices, which is valuable when trips include multiple flight segments and hotel changes in one day (TripIt Help, 2025). The presentation emphasises clarity, with an ordered timeline, maps and attached documents, and the ability to view plans when offline. These design choices support quick retrieval of the next action, for example gate information or hotel check-in details, without navigating separate vendor applications (TripIt Help, 2023). TripIt Pro's flight alerts and airport guidance further strengthen the day-of-travel experience by surfacing time-critical changes promptly and helping the user navigate unfamiliar terminals efficiently, which is particularly useful for tight connections and irregular operations (TripIt Pro, 2025). For Itinera, these strengths highlight the usefulness of automated ingestion where available, clear timeline views for day execution draw attention to information that affects the next step in the plan.

Weaknesses

TripIt does not function as a discovery or planning engine for leisure activities. It documents what the traveller has already arranged rather than suggesting points of interest or constructing a sightseeing schedule. As a result, users must add attractions manually and judge feasibility themselves for a free day in a city, including opening hours and travel time between stops (TripIt Help, 2025). TripIt's feature set also does not include a budget planning module for day itineraries, so travellers cannot set a cost limit and have the app generate an activity plan that stays within that amount, which is a priority for Itinera's scope. Although TripIt supports document uploads and a clean timeline, its primary focus is logistics rather than preference-aware activity sequencing, meaning it does not compute an ordered, single-day tourist route that maximises utility under time and budget constraints (TripIt Help, 2025). Itinera is designed to fill this gap by generating a feasible, timed day plan from user preferences and limits, and by enabling quick re-runs when the user adjusts the budget or available hours.

TripHobo

Overview

TripHobo offers a mobile trip planner that helps users assemble day-wise itineraries with attractions, food venues and bookings in one place. The platform supports creating a customised itinerary for a destination, organising activities into daily schedules, and integrating accommodation and transport to reduce context switching between multiple services during planning (TripHobo, 2018; TripHobo, 2017a). TripHobo also markets a budget calculator that estimates likely costs for elements of a trip, which can guide financial expectations during planning and may help users compare options before committing to reservations (TripHobo, 2017b; TripHobo, 2017c). The mobile app mirrors the web experience and promotes sharing itineraries with friends and family, which assists coordination when others want to review or follow the plan on their devices (TripHobo Mobile, 2018). For short city visits, these capabilities allow a traveller to gather candidate places and arrange them into a structured day without building a schedule from scratch in a spreadsheet.

Strengths

TripHobo's integrated workflow reduces friction, because users can move from selecting attractions to arranging a sequence and referencing suggested durations within a single environment, and can link out to partnered booking services for accommodation and tours when needed (TripHobo, 2018). The system provides breadth of content for destinations, which can be a useful starting point when a traveller is unfamiliar with a city and needs an initial longlist of points of interest. The presence of a budget calculator and cost estimation pages offers high-level guidance on likely spend, which can be valuable during early planning when users are deciding between alternative days or neighbourhoods to focus on (TripHobo, 2017b; TripHobo, 2017c). For Itinera, these strengths suggest design patterns worth adopting, namely concise attraction cards with indicative time on site and price information, and a planning surface that keeps discovery and day structuring close together so users are not forced to jump between tools.

Weaknesses

Public product information emphasises estimates and customisation, but it does not indicate that TripHobo automatically generates a single-day plan that is guaranteed feasible under explicit user-set time and cost limits. In practice, this means travellers may still need to adjust sequences manually if opening hours or inter-point travel times make the day impossible, and they may need to monitor the total cost themselves rather than enforcing a strict cap during generation (TripHobo, 2018). The budget calculator communicates expected spend rather than acting as a binding constraint within an optimiser, which differs from Itinera's goal of computing an itinerary that satisfies time, budget and opening-hour windows by construction (TripHobo, 2017b). Documentation also focuses on adding and arranging items, not on automated constraint checking for each change, which can increase effort for users who want a dependable, timed route for a single day without fine-tuning. Itinera is intended to address these limitations by producing a feasible, timestamped sequence directly from user preferences and constraints, and by allowing rapid recomputation when the user adjusts inputs.

Wanderlog

Overview

Wanderlog is a collaborative itinerary planner for mobile and web that combines reservations management with daily activity planning. Users can forward bookings into a trip and maintain flights and hotels alongside points of interest, notes and attachments, which consolidates logistics and sightseeing in a single workspace (Wanderlog Help, 2024a). A core proposition is group planning, travellers can invite others to view or edit the same trip, which reduces reliance on ad hoc documents and messages when coordinating a shared day in a city (Wanderlog Help, 2023). Wanderlog also includes budgeting tools and expense entry so groups can track spending and compare totals against a stated budget inside the same plan, which is helpful for cost-conscious visitors (Wanderlog Help, 2024b; Wanderlog Help, 2024c). Some capabilities, including offline access to trip plans and downloaded guides, are part of the premium tier, which provides convenience during travel where connectivity is limited (Wanderlog Help, 2023a; Wanderlog Help, 2023b).

Strengths

Wanderlog's collaboration model is a clear differentiator. The ability to share a trip and grant edit permissions enables multiple travellers to add suggestions, rationalise overlaps and converge on a workable plan without maintaining parallel copies, which improves transparency and reduces rework (Wanderlog Help, 2023). The application also supports ingesting reservations by email and presents a day-by-day schedule with a map view, giving users spatial context as they arrange activities and move items between lists and the itinerary, which supports quick sense-checking of sequences (Wanderlog Help, 2024a). Built-in budgeting and expense features provide visibility over cumulative spend and progress against a stated budget, which encourages informed choices while planning food and ticketed attractions (Wanderlog Help, 2024b; Wanderlog Help, 2024c). For Itinera, these strengths underscore the value of clear mapping, simple daily schedules and, where appropriate, controlled collaboration so that companions can understand the plan and propose alternatives without undermining the itinerary's structure.

Weaknesses

Although Wanderlog includes budgeting and route optimisation tools, available documentation does not indicate that it produces an automatically feasible single-day itinerary under explicit user-set time and cost constraints. Users can optimise route order to improve efficiency and can track expenses, however they remain responsible for ensuring the day fits opening hours and finishes within the available time, and for pruning items until the schedule is realistic. This contrasts with Itinera's aim to compute a feasible, timestamped sequence from the outset, given a day window and a budget. Additionally, some useful capabilities, including offline access to the full trip plan and downloaded guides, are limited to premium subscribers, which can reduce utility for travellers who rely on offline use during the day (Wanderlog Help, 2023a; Wanderlog Help, 2023b). Finally, while collaboration is a strength, it can encourage ad hoc edits that make feasibility harder to maintain without automated constraint checks. Itinera is designed to mitigate this by keeping input adjustments intentional, for example budget and preference weights, and by recomputing a plan that remains within constraints.

User Stories and Sprints

Scope Statement

Itinera is a clean and lightweight mobile application that provides tourists with a personalised travel itinerary given their time schedule, cost budget and preferred attractions. We aim to deliver a travel itinerary that maximises the tourist's utility while providing interactive features such as allowing the user to edit their budget and view POI details. We will exclude features such as adding, removing and reordering POIs. The primary goal of Itinera is to assist tourists with the complexity of navigating unfamiliar destinations given limited resources and areas of interest.

Client Approval:

- Date: 03/10/2025
- Method: Email
- Written Acknowledgement: Our project scope statement has been shared with Arindam and Dip (our clients)

Client Approval Evidence:

COMP3900 Project Inbox ×

Jonathan Liu <jonathan.yuangliu@gmail.com> Thu 2 Oct, 12:39 (1 day ago)
to Arindam, Dip

Hi everyone,

Our team has finished the project scope statement which has been shared with you as a Google Doc. I will also provide the statement below.

Project Scope Statement:
Itinera is a clean and lightweight mobile application that provides tourists with a personalised travel itinerary given their time schedule, cost budget and preferred attractions. We aim to deliver a travel itinerary that maximises the tourist's utility while providing interactive features such as allowing the user to edit their budget and view POI details. We will exclude features such as adding, removing and reordering POIs. The primary goal of Itinera is to assist tourists with the complexity of navigating unfamiliar destinations given limited resources and areas of interest.

If you could give us an approval reply for this statement, that would be great! This approval reply is needed to progress onto other stages of the project.

Kind regards,
Jonathan

Jonathan Liu 13:02 (5 hours ago)
Hi team, Just following up on the previous email. Kind regards, Jonathan

---

Arindam Pal 18:28 (4 minutes ago)
to me, Dip

Hi Jonathan,

This looks good to me.

Dip, do you have any comments?

Regards,
Arindam Pal, PhD
Chief Executive Officer and Director
TechSoftX Corporation
Where technology meets software to make positive impacts on human society

•••

Sprint 1 User Stories

User Onboarding and Basic Itinerary Functions

Jira Ticket	User Story	Acceptance Criteria	Points
H13ABANANA-1	As a tourist, I want to have my own profile, so that I can access my saved itineraries.	● Successful login redirects to home page. ● Home page displays saved itineraries.	2
H13ABANANA-2	As a tourist, I want to create a new itinerary, so that I can start planning my trip.	● A “New Itinerary” button is accessible from the profile page. ● Clicking it prompts the user to input date, budget and preference details. ● Clicking “Create trip” generates a new itinerary shell on the profile page.	2
H13ABANANA-3	As a tourist, I want to set my preferred start and end time for the day, so the app can plan within my schedule.	● Time picker allows selection of start and end times.	2
H13ABANANA-4	As a tourist, I want to set my budget, so the itinerary can respect my cost limits.	● User can enter a numeric budget into field.	2

H13ABANANA-5	As a tourist, I want to select my categories/points of interest so the itinerary matches my interests.	● A predefined list of POIs for the destination is present which the user can select from.	3
H13ABANANA-8	As a tourist, I want to delete an itinerary, so that I can remove ones I don't need.	● Each itinerary on the dashboard has a "delete" option. ● Confirming delete removes the itinerary from the dashboard.	2
H13ABANANA-17	As a tourist, I want to receive an itinerary that meets all my requirements and maximises utility so that I can have a good trip.	● Itinerary is generated through a heuristic algorithm and later ● Modeled as an integer linear problem for CPLEX/Gurobi to solve. ● Generated itinerary maximises utility whilst respecting the user-defined constraints and preferences.	13
H13ABANANA-23	As a tourist, I want to view the generated itinerary on a map so that I can see the geographical locations of the POIs.	● POIs are marked on the map with pin icon. ● Pin icons sequentially numbered to represent order of visit. ● Travel route represented by bold path between POIs	8

H13ABANANA-27	As a tourist, I want the system to generate an itinerary that meets my interests, so that I can have a personalised trip.	● User is given categories of interest to select from ● Selected categories will be used to generate a “utility” score for POIs ● Generated itinerary will maximise the utility score while meeting user-defined constraints, otherwise will alert the user if not possible.	5

Itinerary Interactions

Jira Ticket	**User Story**	**Acceptance Criteria**	**Points**
H13ABANANA-7	As a tourist, I want to view my saved itineraries, so that I can revisit them.	● Profile page displays saved itineraries. ● Clicking an itinerary redirects to map view/timeline view page. ● Page displays message if no saved itineraries.	2
H13ABANANA-20	As a tourist, I want the itinerary to update if my budget changes, so that I can stay within my cost limit.	● Editable budget field is present on the itinerary page. ● Itinerary route is recomputed upon editing budget. ● New route and updated statistics displayed on map.	2

H13ABANANA-21	As a tourist, I want the itinerary to update if my time constraint changes, so that I stay within schedule.	● Editable time field is present on the itinerary page. ● Itinerary route is recomputed upon editing time. ● New route and updated statistics displayed on the map.	3
H13ABANANA-22	As a tourist, I want to view details of each POI on the itinerary so that I can plan ahead and preview the day.	● Clicking a POI on the map will trigger a pop up card that displays its details e.g name, arrival time and departure, cost, opening hours. ● Itinerary page displays timeline of itinerary with full POI details.	2
H13ABANANA-6	As a tourist, I want to save the itinerary, so that I can review it in the future.	● “Save Itinerary” button is present in the corner of the map and timeline page. ● Clicking it saves current state of itinerary in database.	2

UI Design

Jira Ticket	User Story	Acceptance Criteria	Points
H13ABANANA-13	As a tourist, I want a clean and simple profile/home page that displays a list of my itineraries so that I can easily navigate the app.	● Profile/ Home page displays all saved itineraries ● Each itinerary shows the correct title, date, image. ● UI is responsive and simple to navigate	3
H13ABANANA-24	As a tourist, I want to view the generated itinerary in a timeline format so that I can conveniently scroll through and view details.	● Itinerary is displayed in vertical timeline format. ● POIs are ordered sequentially by visit order. ● Each POI displays details e.g name, arrival time and departure, cost, opening hours.	5
H13ABANANA-25	As a tourist, I want to view a summary of the statistics of the itinerary so that I can plan ahead.	● Map and timeline page displays total cost, time, duration and utility score of itinerary in corner.	1
H13ABANANA-16	As a tourist, I want a universal navigation bar so that I can easily switch between my profile, map view, and timeline view.	● Navigation bar is present at bottom of app on ALL pages ● Navigation bar link to profile, map and timeline pages.	1

Jira Ticket	User Story	Acceptance Criteria	Points
		● Current active page is highlighted	

Suggestion Algorithm

Jira Ticket	User Story	Acceptance Criteria	Points
H13ABANANA-28	As a tourist, I want the generated itinerary to fall within my cost budget, so that I can enjoy my trip without overspending.	● User can input budget value into field ● Algorithm computes itinerary whose total cost does not exceed the budget ● Total cost of itinerary is displayed in summary shell	5
H13ABANANA-29	As a tourist, I want the generated itinerary to fall within my time schedule, so that I can enjoy my trip without rushing.	● User can set start time and end into time picker ● Algorithm computes itinerary that fits within time window ● Total duration of itinerary is displayed in summary shell	5

H13ABANANA-30	As a tourist, I want the generated itinerary to prioritise certain constraints e.g cost and time so that it reflects my unique travel style.	● Users can adjust a weighting slider for cost and time to indicate how much they care for it. ● Algorithm takes into account the weighting when calculating the utility score and final itinerary.	5
H13ABANANA-26	As a tourist, I want to be notified if certain POIs couldn't be included in the itinerary so that I can adjust my requirements.	● Exclamation icon is present in the corner if computed itinerary excludes POIs. ● Clicking the icon displays further details on why POI was excluded e.g "Not enough time", "Exceeds budget".	3

User story allocation for sprint 1

H13ABANA Sprint 1 2 Oct – 16 Oct (9 work items) 24 13 0 Complete sprint

- come up with basic algorithm - setup environment - basic UI

H13ABANANA-1	User profile	TO DO	2
H13ABANANA-2	User clicks itinerary creation	TO DO	2
H13ABANANA-3	Setting travel times	TO DO	2
H13ABANANA-4	Budget/cost constraint on travel itinerary	TO DO	2
H13ABANANA-5	Categories/POIs for itinerary	TO DO	3
H13ABANANA-8	Delete saved itinerary	TO DO	2
H13ABANANA-17	Itinerary generation algorithm	IN PROGRES...	13
H13ABANANA-23	View POI and itinerary on map	TO DO	8
H13ABANANA-27	Calculate utility score	TO DO	3

Product Backlog

Backlog (13 work items) 39 0 0 Create sprint

H13ABANANA-6	Saving itineraries	TO DO	2
H13ABANANA-7	Viewing saved itineraries	TO DO	2
H13ABANANA-16	Universal navigation bar	TO DO	1
H13ABANANA-13	Clean and simple UI	TO DO	3
H13ABANANA-20	Budget constraint update itinerary	TO DO	2
H13ABANANA-21	Time constraint update itinerary	TO DO	3
H13ABANANA-22	View POI detail	TO DO	2
H13ABANANA-24	View itinerary timeline format	TO DO	5
H13ABANANA-25	Display summary statistics of itinerary	TO DO	1
H13ABANANA-26	Notice of POIs that could not be included	TO DO	3
H13ABANANA-28	Itinerary meets budget constraint	TO DO	5
H13ABANANA-29	Itinerary meets time constraint	TO DO	5
H13ABANANA-30	Itinerary prioritises time/cost	TO DO	5

+ Create

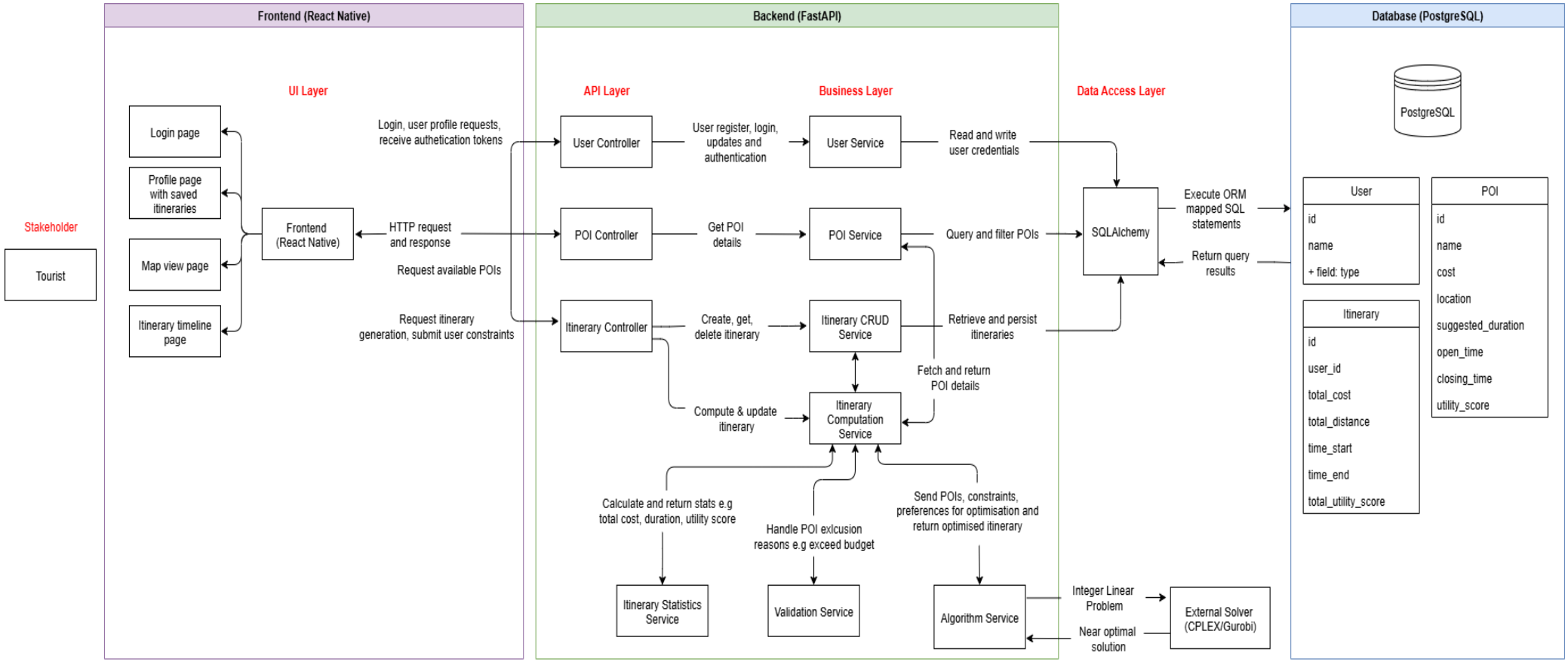
Sprint Timeline and Milestones

Sprints	**Dates**	**TODO and Milestones**
Sprint 1	Start date: W3, Wed 1st Oct End date: W5, Thursday 16th Oct	1. Backend and frontend containers running via Docker. (6th Oct) 2. Brainstorming and writing potential algorithms on paper. (7th Oct) 3. Design drafted algorithm following budget, time and POI constraints. (13th Oct) 4. Map with POIs viewable on app. Users can click on POIs, and user profile implemented. (16th Oct) **Progressive Demo A: 16 Oct**
Sprint 2	Start date: W5 Thursday 16th Oct End date: W8 Thursday 6th Nov	1. Users can input constraints and the itinerary is successfully generated. The algorithm from sprint 1 is implemented. (25th Oct) 2. Testing with small sample datasets added to the app. (25th Oct) 3. Itinerary generation algorithm is further optimised. (30th Oct) 4. Backend fully integrated with frontend for all itinerary generation. (6th Nov) 5. The UI is clean and functional across app screens. (6th Nov) **Progressive Demo B: 6 Nov**

Sprint 3	Start date: W8 Thursday 6th Nov End date: W10 Thursday 20th Nov	1. The algorithm is polished and tested with bigger datasets to ensure a fast and robust system. (12th Nov) 2. Bug fixes, all user stories completed (14th Nov) 3. Final UI sweep. (19th Nov) 4. Documentation up to date and completed. (19th Nov) 5. Product ready for demo. (19th Nov) **Final Demo: 20 Nov**

Technical Design

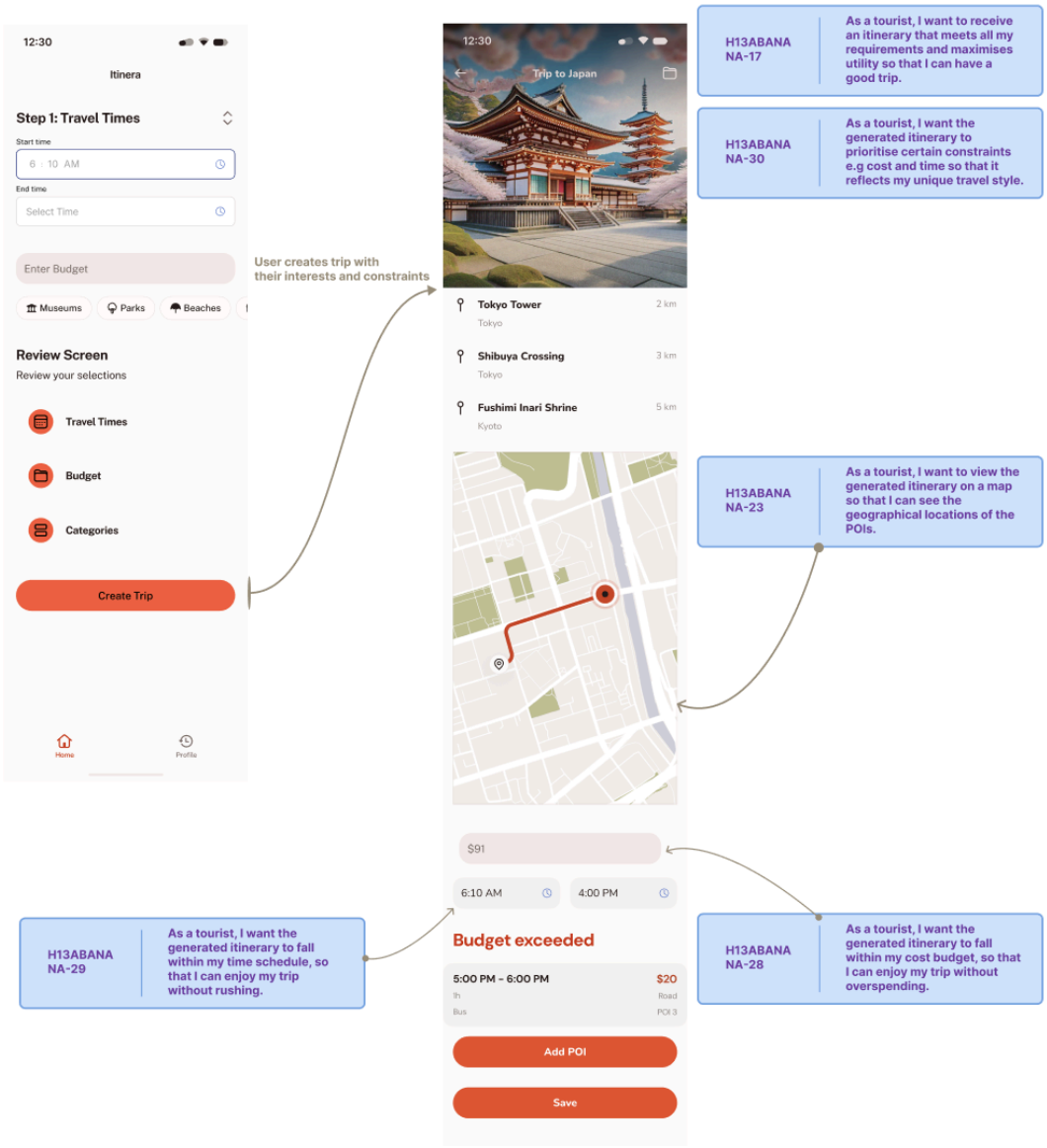
System Architecture Design



Interface Storyboarding

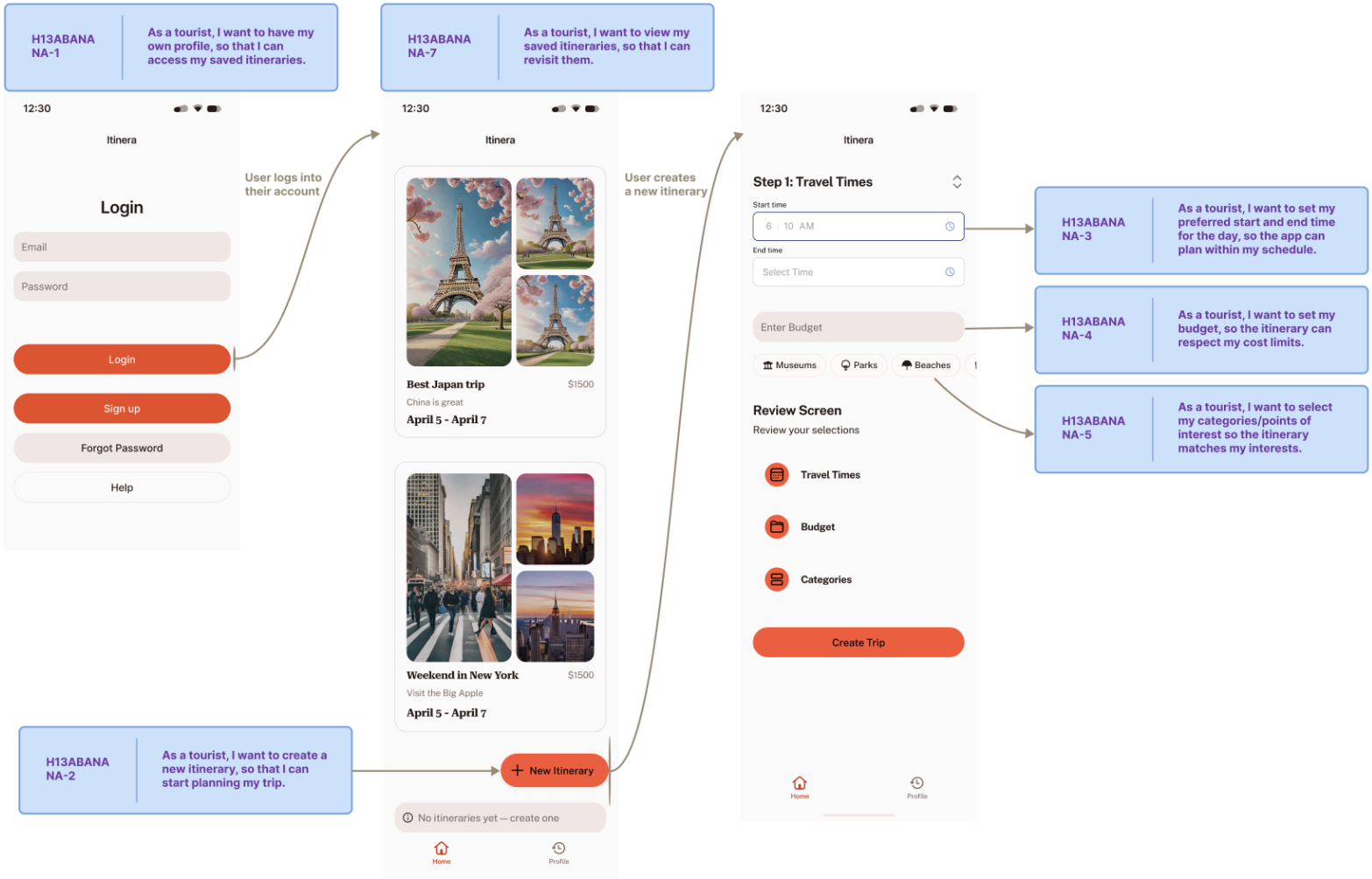
Storyboard 1: This storyboard demonstrates three key functionalities:

- H13ABANANA-17/30: Itinerary Generation
  - The system generates a trip plan that aligns with the tourist's constraints (time, budget, POIs), maximising utility.
- H13ABANANA-23: Map View
  - The itinerary is displayed on a map with POIs and distances, helping the tourist visualise their travel route.
- H13ABANANA-28/29: Editing Constraint
  - If the plan exceeds the budget or time window, the system highlights the issue (e.g., "Budget exceeded") and provides options to adjust POIs.



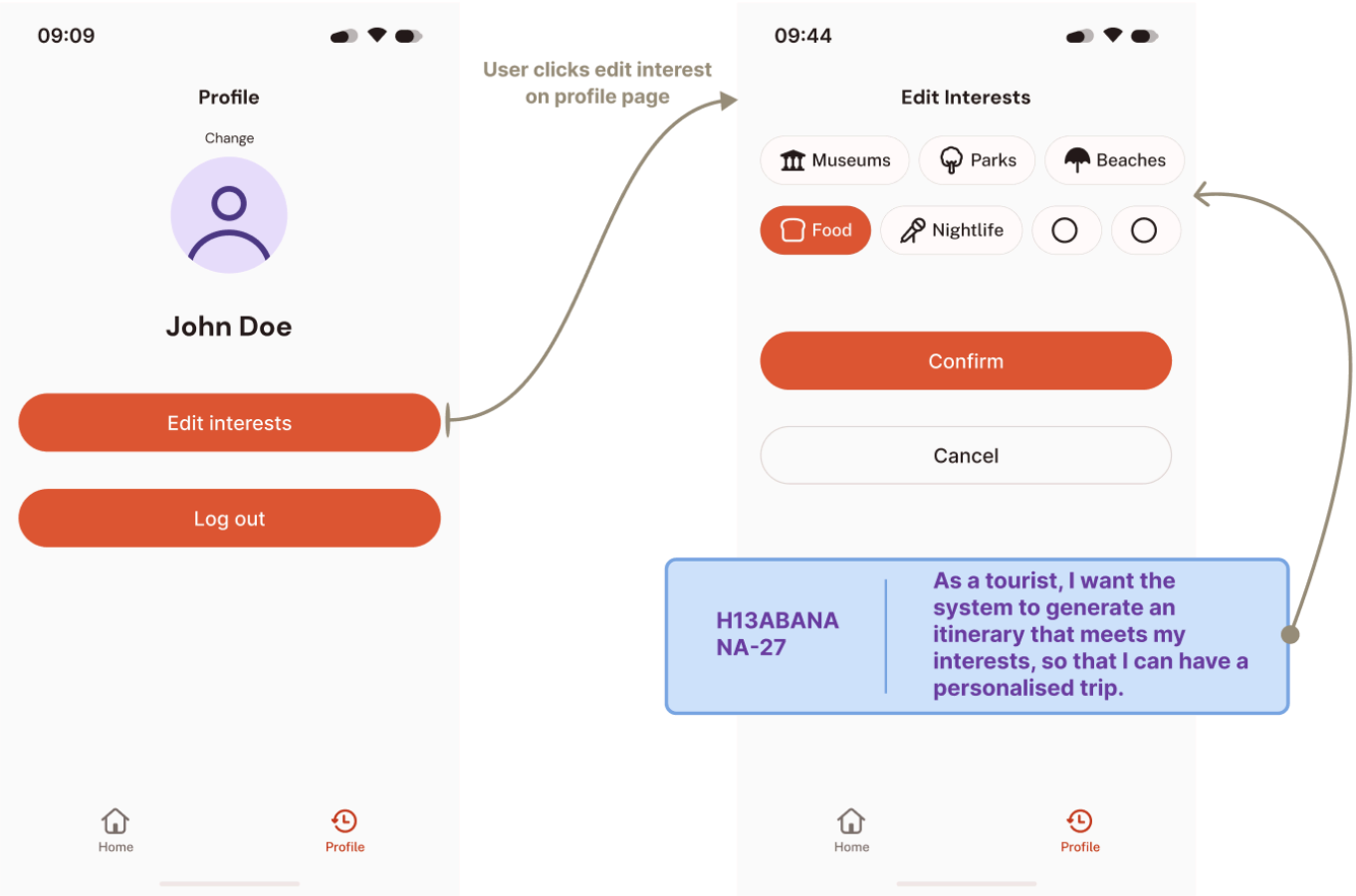
Storyboard 2: This storyboard demonstrates two key functionalities:

- H13ABANANA-1/7: Logging in and viewing saved itineraries
  - The user can either sign up or login to see their previously saved itineraries on the home page.
- H13ABANANA-2/3/4/5: Creating an itinerary
  - The user can create a new itinerary and put in time, cost and category constraints so a personalised itinerary is generated.



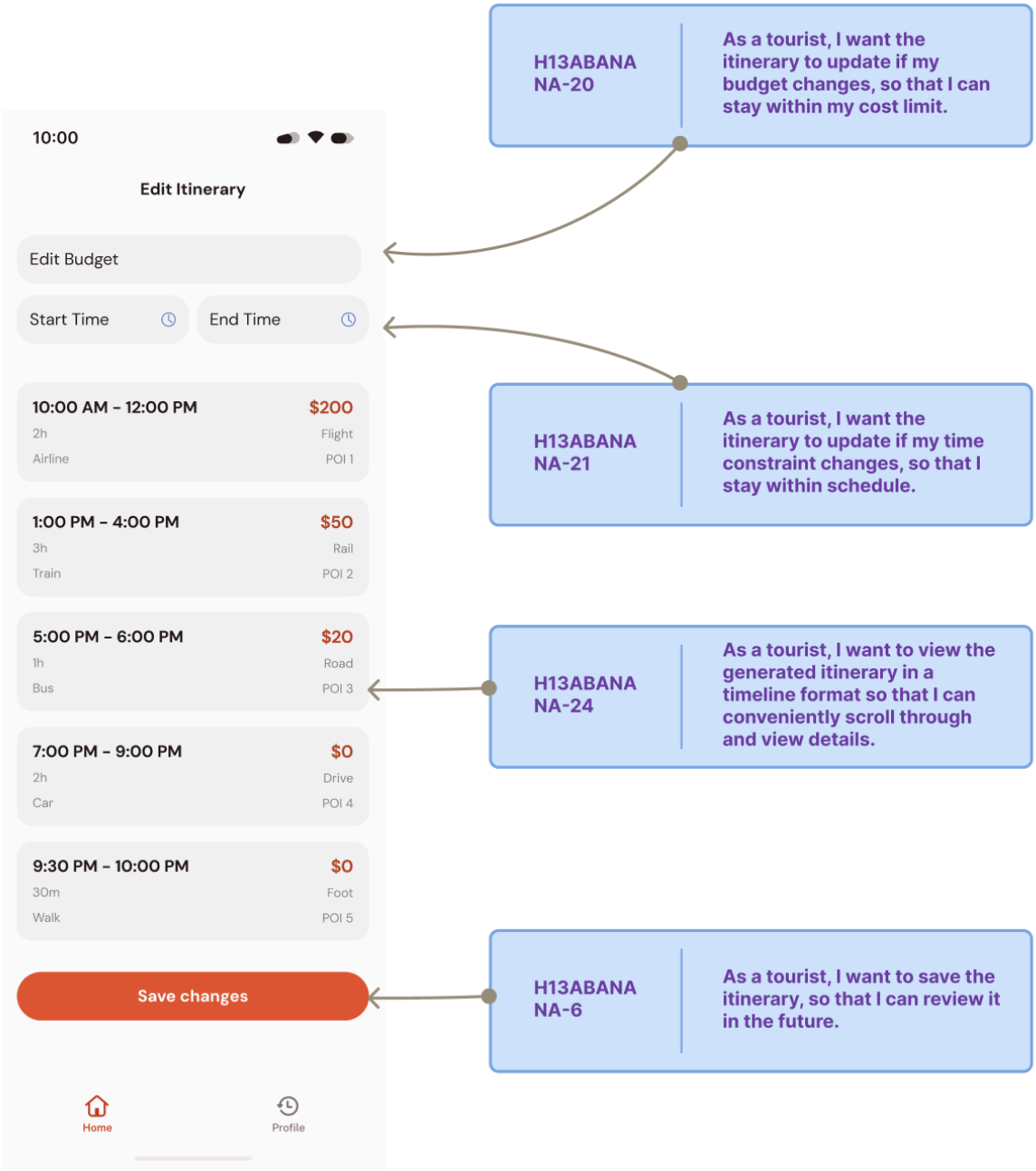
Storyboard 3: This storyboard demonstrates one key functionality:

- H13A-BANANA-27: Edit Interests
    - The user selects categories of interest that will be used to calculate the utility score of POIs.



Storyboard 4: This storyboard demonstrates 3 key functionalities:

- H13A-BANANA-20/21: Update Changes
    - The itinerary will update in accordance with the budget and time changes set by the user.
- H13A-BANANA-24: Timeline View
    - The itinerary is displayed as a timeline with POIs, times, costs and mode of transport which helps the tourist visualise their travel route.
- H13A-BANANA-24: Save Changes
    - The user can save changes for further reference in the future.



Design Justifications

The overall problem is generating a personalised, optimal *and* efficient travel itinerary that maximises utility for a tourist. This is challenging due to many contributing factors such as including time and budget constraints, personal interests and travel feasibility. We have broken this large problem down into the following subproblems:

1. **Utility score definition**

Assigning utility scores to each POI is crucial to measure its benefit to the tourist. This score must be meaningfully calculated in a way that considers all the tourists's constraints. If it is poorly defined, the generated itinerary will not be personalised for the tourist's preferences.

We plan to solve this by defining a composite utility score that is calculated as a weighted sum or product of several factors.The total cost of visiting a POI e.g entry fee, booking fee will be considered and POIs that fall within the user's budget will be favoured. Additionally, the average duration spent at a POI and travel time to reach it will be factored. The tourist's interests will also be reflected with certain POIs being boosted through multipliers when matching user preferences.

We initially considered using static utility scores for each POI that are fetched from public reviews and numerical ratings online. However, this would not produce a personalised itinerary. Each individual has different interests as well as time and budget constraints that affect the value visiting a POI will bring. Hence, this is why we chose our solution which incorporates time, cost and categories of interest into the utility score calculation. This approach makes our application more adaptive and superior to existing systems like TripHobo and Wanderlog which often generate itineraries that are not guaranteed to respect the user's time and cost constraint.

A limitation of this design is that it does not take into account the tourist's preferences *within* categories of interest, thus making it not truly personalised. We planned to mitigate this by allowing users to rank their categories of interest before the itinerary generation, however, this additional step may make the user flow too complex. We also considered fetching the user's preferences once on sign up and storing it in the profile, however, this introduces complexity in the database layer. Additionally, it would not be intuitive to change categories in the profile if the user plans to generate multiple itineraries for different purposes.

The most likely solution we will adopt is to seek user feedback through allowing users to rate POIs or select like/dislike after completing an itinerary. This approach is minimally disruptive to the user flow and enables the system to gradually learn and adapt to individual preferences over time.

2. **Weighting Time and Cost Constraints**

When multiple itineraries are feasible, the system should choose the one that reflects the tourist's preference for staying within time or cost budget. For example, some individuals have greater spending capacity and do not care for choosing a more expensive mode of transportation, while others value minimising cost.

We plan to solve this problem by introducing a slider that allows users to express the extent to which they prefer minimising time or cost. The middle position represents no preference and the distance from the center position indicates the degree of preference. The algorithm will subsequently reflect bias towards one factor.

An alternative solution for preference weighting was to have three buttons for the user:

- Prioritise time
- Neutral
- Prioritise cost

However, this solution only offers three discrete options. A slider provides continuous weights, resulting in more nuanced outcomes and improved personalisation for the user.

3. **Travel methods between POIs**

The travel method between POIs greatly impacts the enjoyment of an itinerary. Without accurate travel modelling, itineraries may look achievable on paper but are impractical in reality. For example, a route may suggest several hours of walking or use of a bicycle that the tourist does not have access to.

We plan to solve this problem by scraping and aggregating accessible transport options for each POI pair. Travel times will be estimated using average walking speeds, public transport intervals or typical travel durations. This solution offers greater functionality over existing systems where transport is not accounted for at all, leaving the user responsible for making arrangements themselves.

However, a limitation of this design is that it may not reflect real time traffic conditions, public transport delays or other factors that make a mode of transport infeasible. This can be mitigated by periodically scraping data so that estimates are up to date or integrating real time APIs like Google Maps.

4. **Interactivity**

Sometimes, it may be impossible to include all the user's preferred POIs due to time or budget limitations. A well designed system must make this transparent to the user and allow meaningful interaction tools for them to adjust their constraints accordingly.

We plan to achieve this by clearly displaying excluded POIs alongside the reason e.g "exceeded budget" or "not enough time". Furthermore, we will have editable fields for budget and time which allows the user to adjust their constraints and immediately generate a new itinerary. This ensures the user remains in control of their trip and allows them to iteratively refine the itinerary until they are satisfied. This differs from other applications that only offer pre-defined itineraries with no scope for changes.

We believe our chosen design offers the greatest flexibility and fosters user trust by providing both explicit feedback and interaction, thus striking a balance between autonomy and usability.

Functionality Mapping

Functionality	User Stories	Justification
Profile	1	The user profile allows the user to view previously saved itineraries, allowing them quick access to these itineraries and improving ease-of-use.
Creating and deleting itineraries	2, 8	Allows the user to create a new itinerary when needed (2) or delete an itinerary from the user's saved itineraries list when no longer needed (8).
Viewing and editing existing itineraries	3, 4, 5, 6, 7, 20, 21	This functionality allows the user to view previously saved itineraries (7), as well as customise these itineraries such as setting travel dates (3) and a budget (4). The user can also select categories/points of interest (5), a more personalised experience. Furthermore, saving an itinerary (6) allows the user to continue where they left off. They can also edit a saved itinerary (20, 21) if their requirements suddenly change.
UI Navigation	13, 16	Allows UI flow for the user to use the mobile application: home page and saved itineraries page to see a list of saved itineraries (13), and a navigation bar to navigate to different pages (16).
Recommendation	17, 27, 28, 29, 30	Provides recommended itineraries

system		based on user preferences. Generate an itinerary that aligns within user-defined requirements of time (28) and budget (17, 29) and selected categories of interest with a specified order of points of interest to maximise utility (27, 30).
Map and Functionality	22, 23, 24, 25, 26	The user will be able to view and select POIs on the map to see its details (22), as well as the generated itinerary on the map (23). Displays generated itinerary in a timeline format (24) and statistics such as cost, time and duration (25), and which POIs couldn't be included in the itinerary (26).

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Client Communication Evidence:

Jonathan Liu <jonathan.yuangliu@gmail.com>
to arindamp, dip

Thu 25 Sept, 13:43 (4 days ago)

Project:
A Fast, Scalable, And Optimal Personalised Travel Itinerary Planning System
(COMP3900)

Dear Arindam Pal and Dip Guha,

We hope this message finds you well. We are writing to introduce ourselves as the student team assigned to your project titled "A Fast, Scalable, And Optimal Personalised Travel Itinerary Planning System" as part of the UNSW COMP3900 Capstone Project course.

We are excited about the opportunity to collaborate with you this term. Our team consists of the following members:

- Jonathan Yuang Liu
Degree: Bachelor of Computer Science And Commerce
Skills: Mixture of Backend and Frontend + Communication

- Sydney Xu
Degree: Bachelor of Computer Science
Skills: Backend + Databases

- Reuben Cuasay
Degree: Bachelor of Actuarial Studies and Computer Science
Skills: Databases + Machine Learning

- Daniel Liu
Degree: Bachelor of Computer Science
Skills: Backend + Databases

- Zimo Luo
Degree: Bachelor of Computer Science
Skills: Backend

- Leo Xu
Degree: Bachelor of Computer Science
Skills: Frontend

To streamline our communication, we have nominated Jonathan as our Product Owner, who will serve as the main point of contact between you and our team. We would like to propose scheduling a first meeting with you during Week 2 (or Week 3 Monday) at your convenience.

The aim of the meeting would be to:
- Clarify project requirements and goals
- Discuss scope, deliverables, and expectations
- Establish preferred communication channels and meeting cadence
- Understand any co-clients or stakeholders involved in the project

Please let us know your availability this week or next Monday and preferred platform for this meeting (e.g., Microsoft Teams, Zoom, Google Meet, etc.).

We have started reviewing the project brief and will come prepared with questions and discussion points. We're eager to learn from you and help deliver a valuable outcome by the end of the term.

Looking forward to working with you!

Warm regards,

Arindam Pal
to me, Dip
Fri 26 Sept, 16:54 (3 days ago)

Dear Jonathan,

We can meet at 3 pm AEST on 29th September.

Dip, does that work for you?

Jonathan Liu <jonathan.yuangliu@gmail.com>
to Arindam, Dip
Fri 26 Sept, 17:50 (3 days ago)

Dear Arindam and Dip,

That would be perfect!

Kind regards,
Jonathan

Arindam Pal
to me, Dip
Fri 26 Sept, 17:57 (3 days ago)

Please send us a calendar invite.

Jonathan Liu <jonathan.yuangliu@gmail.com>
to Arindam, Dip
26 Sept 2025, 18:09 (3 days ago)

Hi team,

You're invited to COMP3900 Client Meeting
Monday, 29 September 2025
3:00 pm - 4:30 pm (AEST)
https://teams.microsoft.com/meet/4763471521128?p=BmPFAm5j5nX9SdZvDZ
Tap on the link or paste it in a browser to join.

Here is the invite link to the meeting. Looking forward to meeting you both!

Kind regards,
Jonathan

Jonathan Liu <jonathan.yuangliu@gmail.com>
to Arindam, Dip
23:15 (12 minutes ago)

Hi team,

Here is the meeting link for next week. Keen to see you all!

You're invited to COMP3900 Client Meeting Monday, 6 October 2025 3:00 pm - 4:00 pm (AEST) https://teams.microsoft.com/meet/4254759422745?p=SoRLwJ6C7FfMjdwgJx Tap on the link or paste it in a browser to join.

Jonathan Liu <jonathan.yuangliu@gmail.com>
to Arindam, Dip
23:26 (0 minutes ago)

Hi team,

Also, I forgot to mention in the previous email that as requested, I have shared our project scope statement with you for review.

Kind regards,

Jonathan